

NEW CALCULATIONS OF PHOTODISSOCIATION CROSS-SECTIONS IN THE O₂ SCHUMANN-RUNGE SYSTEM

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ABSTRACT

The large amount of new spectroscopic information that has become available in recent years makes recalculation of atomic oxygen production rates by photodissociation in the Schumann-Runge band region necessary. The results of such a calculation indicate that, on the whole, the new spectroscopic data do not give rise to large changes from the existing recommendations of the World Meteorological Organisation. However they do indicate higher atmospheric transmission in the region 194 to 200 nm, as observed by recent balloon experiments. Parameterisation of the results is also considered and a method based upon O₂ column density and local atmospheric temperature is presented. This parameterisation is accurate to better than 5% for a wide range of solar zenith angles and atmospheric conditions.

Keywords: Schumann-Runge, Oxygen cross-sections, Oxygen photodissociation.

1. INTRODUCTION

Photodissociation of O₂ in the region of the Schumann-Runge (SR) bands (175-200 nm) is the dominant source of odd oxygen in the mesosphere and an important contributor to the stratospheric production. Its proper representation in photochemical models of the middle atmosphere is therefore an important consideration. However the complex band structure prohibits a detailed treatment in such models because of the computational expense involved. Accordingly a simplified parameterisation is necessary. This usually involves dividing the region into a manageable number of spectral intervals which in standard modelling practice are 500 cm⁻¹ wide and 17 in number (Ref. 1). Effective cross-sections and transmissions for each interval as a function of penetration depth into some standard atmosphere are then calculated and tabulated. These can be used directly as suggested by Frederick in the World Meteorological Organisation's (WMO's) 1986 report on atmospheric ozone (Ref. 2) or be fitted with polynomial functions as done by Allen and Frederick (Ref. 3). The latter authors have also looked at the effect of solar zenith angle. This effect arises as a result of the temperature dependence of the absorption cross-section and the fact that the temperature profile as a function of absorption path length is dependent on the direction of penetration.

Since these studies were carried out a large amount of new spectroscopic information has become available regarding line positions, line widths and band oscillator strengths (Refs. 4, 5, 6). These new data led Murtagh (Ref. 7) to perform new detailed calculations. As a result of this work the author realised that existing parameterisations were not applicable to arbitrary model atmospheres without introducing errors on the order of 20% in the

photodissociation rates. This paper will briefly review the new calculations and will present a scheme for parameterising the SR band photodissociation that is accurate for a variety of zenith angles and atmospheric conditions.

2. THE CALCULATIONS

2.1 The absorption spectra

Absorption cross-sections were calculated with a resolution of 0.02 cm⁻¹ for the 17 spectral regions. Each of the over 7000 individual lines encompassed by bands with $v' = 0-19$, $v'' = 0-1$ and $N' = 1, 3, 5, \dots, 31$ was allowed to contribute to every spectral region since the quasi-continuum formed by the overlapping wings of the Voigt line profiles is non-negligible. The computations were made for four temperatures (150, 200, 250, 300 K). Figure 1 shows a section of the absorption cross-section in the region between 51 000 and 51 500 cm⁻¹ for the two extremes of temperature. The presence of the 6-1 band at the higher temperature should be noted. This is a result of the thermal population of the $v'' = 1$ state although its prominence, since the fractional population is very low, arises from the order of magnitude larger band oscillator strengths for transitions from this state (Ref. 7).

2.1.1 The underlying continua. Apart from the quasi-continuum formed by the overlapping of the line wings there are two other main continua underlying the SR bands. These are the Herzberg continuum which dominates at the long-wavelength end and the temperature dependent SR continuum which contributes strongly at wavelengths below 185 nm. In this work the formula of Nicolet and Kennes (Ref. 8) has been used to describe the wavelength dependence of the Herzberg continuum. The strong temperature dependence of the SR continuum in the region arises because the absorption relies upon thermal population of rotational and vibrational states above ground state ($v'' = 0$, $N'' = 1$). The absorption threshold for the ground state itself is 174.5 nm and therefore it does not contribute. A model for the absorption due to this process was constructed using relevant data from Refs. 7, 9 and 10. Full details of this and the other calculations have been given in Ref 11.

2.2 Atmospheric transmission and effective cross-sections.

The calculation of transmittances and effective cross-sections involves following the absorption of solar radiation down into the atmosphere at full resolution and evaluation of the relevant quantities at each atmospheric level. For these calculations the atmosphere from 120 to 20 km was divided into layers 5 km thick. Densities and temperatures were taken from the US Standard Atmosphere (Ref. 12) for the standard case. Computations were

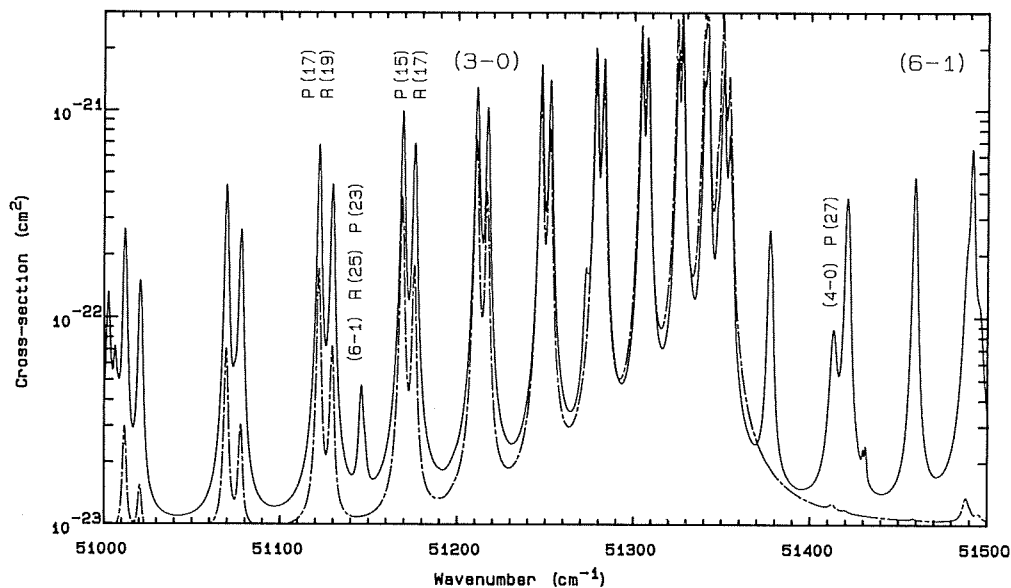


Figure 1. Variation of the absorption cross-section in spectral region 12 as the temperature is raised from 150 (dashed line) to 300 K (solid line). Note the appearance of the (6-1) band.

done for five solar zenith angles (0° , 30° , 60° , 75° and 85°). The path lengths in each layer were calculated taking the earth's curvature into account. This only becomes important for the larger solar zenith angles where the path enhancement factor, $\sec(\chi)$, in the plane atmosphere approximation, is no longer adequate. Murtagh (Ref. 11) has compared the results of the new calculations with stratospheric balloon measurements of atmospheric transmission and found them to be in good agreement. This represents an improvement over the current WMO recommendations. Figure 2 shows the variation of effective cross-section as a function of height and solar zenith angle for regions 10, 11 and 12 (numbering of the region follows Refs. 2 and 11). Obviously this representation highlights the zenith angle effect since, for a given height, each zenith angle corresponds to a different optical depth. Figure 3 is a more honest representation. In this figure the effective cross-sections are plotted against the column density of O_2 along the path and the curves lie more closely together. However the differences are sufficiently great to indicate that a simple parameterisation in terms of only O_2 column density will lead to

significant errors. Further, as pointed out in Ref. 11, even with a fixed illumination direction (overhead sun) differences of around 20% in dissociation rates can occur for a given O_2 column density level, as a result of seasonal changes in the atmospheric temperature profile. This is particularly true at high latitudes where such seasonal changes are most pronounced. Figure 4 illustrates this by showing the calculated dissociation rate constant for overhead sun conditions for two atmospheres typical of summer and winter conditions at $65^\circ N$. Of course overhead sun conditions at $65^\circ N$ are physically impossible but introducing a more realistic zenith angle nearly shifts the regions of discrepancy to other O_2 column densities. It is important to note that the regions of discrepancy occur where the respective atmospheres show substantially increased temperatures compared to the other; that is, in the summer stratopause region and the winter mesopause region. This observation and a closer study of the zenith angle dependencies lead to the conclusion that, it is not so much the temperature profile along the absorption path, as intimated by (Ref. 2) but the local temperature that determines the effective cross-

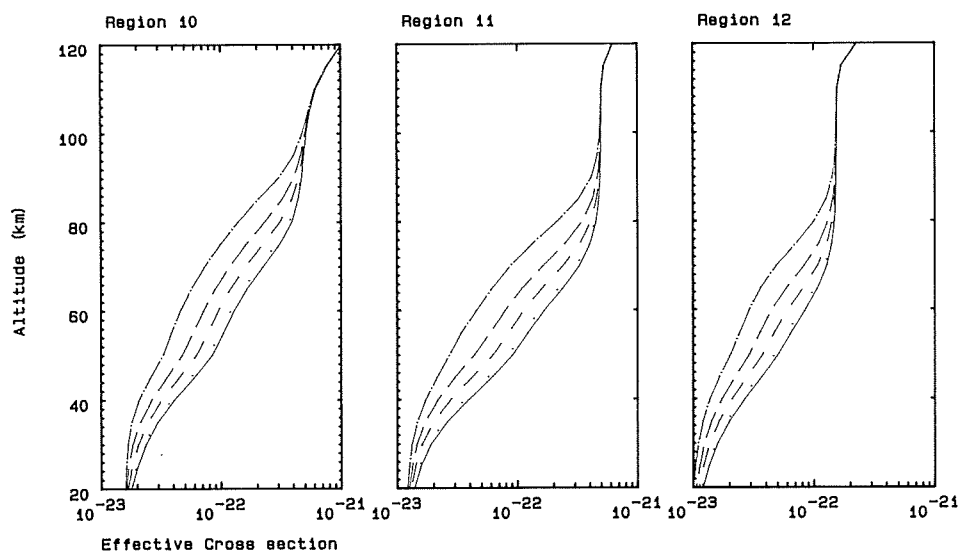


Figure 2. Altitude profiles of effective cross-sections in regions 10, 11 and 12 for solar zenith angles of 0° , 30° , 60° , 75° and 85° .

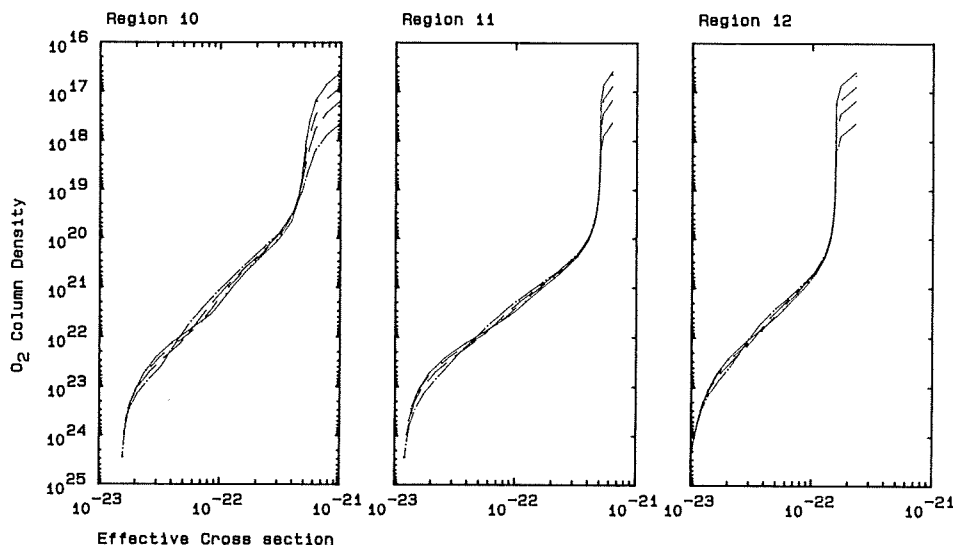


Figure 3. Effective cross-sections for regions 10, 11 and 12 as a function of O_2 column density and solar zenith angle.

section even at O_2 column densities where opacity effects are considerable. This is a consequence of the fact that the increase in absorption cross-section is a result of new spectral features, from the $v^* = 1$ level, appearing in parts of the spectrum that have not been substantially extinguished along the path to the region of increased temperature. Absorption in the warm thermosphere is insufficient to prevent the increasing $v^* = 1$ bands from contributing to the effective cross-section in the warm stratopause area for most of the spectral regions.

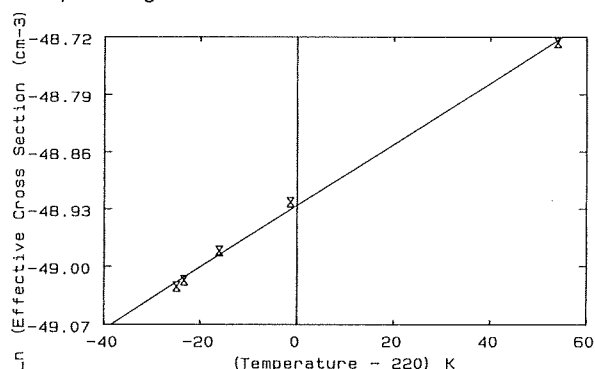


Figure 4. Variation of the logarithm of the effective cross-section with local temperature at an O_2 column density of $1 \times 10^{19} \text{ cm}^{-2}$ in region 10. The straight line represents the result of a linear fit.

3. PARAMETERISATION

A parameterisation must reproduce, as faithfully as possible, the results of an exact calculation for the specific conditions of the model. Since the atmospheric parameters in current models involve a range of seasonal, latitudinal as well as solar zenith angle variation, the parameterisation must be able to cope with these. As seen above this rules out a simple parameterisation in terms of O_2 column density alone or even a more complicated one where the solar zenith angle is treated as in Ref. 2. However the observation that the local temperature is important in determining the effective cross-section, even at large opacities, offers hope for a relatively simple scheme that will take care of both zenith angle and seasonal effects. With this in mind we looked at the correlation of effective

cross-section with local temperature at fixed O_2 column densities for the five data sets corresponding to varying zenith angle. As illustrated in Figure 4 we found a dependence of the logarithm of the cross-section on the local temperature that was near linear for the majority of O_2 column densities. Accordingly, for a given O_2 column density the effective cross-section can be expressed as

$$\ln\{\sigma_{\text{eff}}(N(O_2))\} = A(N(O_2)) [T(z) - 220] + B(N(O_2)) \quad (1)$$

where $N(O_2)$ is the column density of O_2 along the absorbing path and $T(z)$ is the temperature at the corresponding altitude in the atmosphere. As a next step the parameters A and B were calculated for O_2 column densities with $30 < \ln\{N(O_2)\} < 56$ in steps of 0.5 for each spectral region.

Figure 5 shows the variation of A and B with O_2 column density for region 10. As can be seen both parameters have a reasonably smooth variation with $\ln(N(O_2))$ and should be suitable for polynomial expansion. The corresponding curves for each spectral region were fitted as a function of $\ln(N(O_2))$ using a Chebyshev polynomial expansion of order 20. The reason for choosing Chebyshev polynomials was that this type of expansion is easily truncated to any desired number of terms without having to refit. This is a result of the so called minimax property of Chebyshev polynomials. Visual inspection of the fitted curves indicated that no more than 10 or 12 terms are needed to obtain an excellent fit. This is further substantiated in Table 1 where the maximum error in the computed cross-sections for each region is given as a function of the number of terms in the expansions for the A and B coefficients in equation (1). It is clear that even with as few as 7 terms the maximum error in the cross-section is less than 8% for all regions and that the average error is much less than this. The fits are based only on full calculations at different zenith angles so a good test of the parameterisation is to apply it to the calculations carried out for different atmospheric temperature profiles. Table 2 lists the maximum errors in the calculated cross-sections for the summer and winter atmospheres mentioned above. The results are very encouraging.

4. CONCLUSIONS

The technique of parameterising the effective cross-sections in terms of the local temperature as well as the O_2 column density has worked well. It allows for both zenith angle effects and varying atmospheric conditions. With as few as 7 terms in the expansions for A and B the maximum error in the effective cross-sections is less than 8% which is better than that achieved in Ref. 2 with a similar computational level. Table 3 gives the first 7 terms in each

Table 1.

Maximum percentage error in the parameterised effective cross-section and the altitude of occurrence as a function of the number of terms in the expansion.

Region	Number of Coefficients in the expansion.													
	5		7		9		11		13		15		20	
	%	Z (km)	%	Z (km)	%	Z (km)	%	Z (km)	%	Z (km)	%	Z (km)	%	Z (km)
1	4.67	75.0	3.49	95.0	1.60	95.0	0.87	75.0	0.66	80.0	0.14	85.0	0.27	85.0
2	0.95	65.0	-5.30	65.0	-2.73	65.0	-3.64	65.0	-4.17	65.0	-3.15	65.0	-1.12	65.0
3	5.11	75.0	-8.12	60.0	-4.06	60.0	-4.03	60.0	-3.46	60.0	-1.24	60.0	1.82	65.0
4	5.53	65.0	-7.07	65.0	-5.51	65.0	-3.75	65.0	-2.71	65.0	-1.85	65.0	-1.45	65.0
5	15.19	55.0	7.24	90.0	4.21	90.0	1.62	60.0	0.82	60.0	0.99	55.0	-0.03	55.0
6	17.30	50.0	5.95	85.0	5.01	85.0	2.82	85.0	0.79	65.0	-1.33	50.0	-1.87	50.0
7	10.66	65.0	-2.64	50.0	0.35	50.0	2.31	50.0	1.27	50.0	-1.01	50.0	-2.82	50.0
8	18.33	45.0	3.61	45.0	3.90	45.0	4.45	45.0	2.31	45.0	1.34	45.0	0.22	45.0
9	17.47	40.0	7.56	70.0	6.06	70.0	4.43	40.0	3.34	40.0	2.92	40.0	2.00	40.0
10	8.94	65.0	7.33	65.0	5.67	70.0	4.46	40.0	3.99	45.0	2.89	45.0	2.00	40.0
11	23.78	40.0	5.07	40.0	4.36	65.0	3.40	65.0	3.59	40.0	3.52	40.0	1.68	40.0
12	14.07	40.0	3.73	55.0	2.51	55.0	2.36	55.0	1.47	60.0	1.35	40.0	1.04	40.0
13	9.22	30.0	5.54	55.0	0.76	30.0	2.85	55.0	2.08	55.0	0.95	40.0	0.98	40.0
14	12.34	25.0	1.74	40.0	1.01	30.0	2.49	45.0	0.77	40.0	1.33	40.0	0.80	40.0
15	6.87	25.0	0.72	35.0	0.77	30.0	0.96	45.0	0.33	40.0	0.63	40.0	0.30	35.0
16	2.91	25.0	-0.12	25.0	0.45	25.0	0.35	45.0	0.14	30.0	0.15	40.0	0.11	35.0
17	0.65	25.0	-0.11	25.0	0.11	25.0	0.17	45.0	0.05	30.0	0.05	40.0	0.05	35.0

expansion for each region. The ordinary polynomial equivalents are available from the author if required although this is not recommended. However for accuracy it is preferable to use the Chebeshev expansion. Routines for evaluating Chebeshev polynomials are available in most numerical libraries (See for example Ref. 12).

Table 2.
Maximum percentage error in the parameterised effective cross-sections and the altitude of occurrence for Summer and Winter Atmospheres (7 coefficients).

Region	Summer		Winter	
	%	Z (km)	%	Z (km)
1	4.37	80.0	4.76	90.0
2	-3.23	60.0	-5.09	60.0
3	7.20	85.0	7.36	80.0
4	-6.01	60.0	7.60	80.0
5	0.22	55.0	7.75	85.0
6	0.18	50.0	8.00	75.0
7	-5.11	50.0	7.66	70.0
8	5.24	45.0	5.44	45.0
9	5.49	35.0	8.34	65.0
10	6.86	65.0	10.00	60.0
11	6.22	35.0	2.38	35.0
12	3.93	40.0	5.45	45.0
13	-3.67	25.0	6.85	45.0
14	1.61	35.0	-1.57	25.0
15	0.70	30.0	-0.54	25.0
16	-0.01	25.0	-0.18	20.0
17	-0.10	25.0	-0.13	20.0

REFERENCES

1. Ackerman M 1971, *Mesospheric models and related experiments* Ed. G Fiocco, Dordrecht, D Reidel Publ. Co.
2. Frederick J E 1986 *Atmospheric Ozone 1985: Assessment of Our Understanding of the Processes Controlling its Present Distribution and Change*, Rep 16, Ch 7. WMO Global Ozone Res. and Monit. Proj., Geneva.
3. Allen N and Frederick J E 1982, Effective photodissociation cross-sections for molecular oxygen and nitric oxide in the Schumann-Runge bands. *J Atmos. Sci.* **39**, 2066.

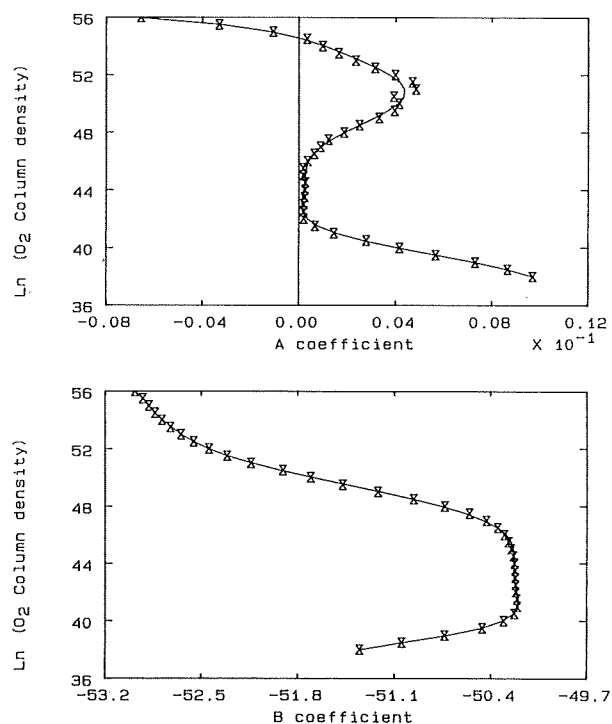


Figure 5. Variation of the linear regression coefficients A and B with O_2 column density for region 12. The solid lines are the Chebeshev polynomial fits with 12 terms in the expansion

Table 3.
The first seven Chebeshev polynomial coefficients for the parameters A and B

Region		c_0	c_1	c_2	c_3	c_4	c_5	c_6
1	A*	-40.745582	-5.407035	-13.311687	12.475974	1.808109	1.655839	3.616947
	B	-92.020912	-2.880716	0.622445	0.379215	-0.190632	-0.103796	0.079266
2	A	49.101012	44.111386	-1.180357	15.720174	2.520101	1.663620	-0.706379
	B	-93.586349	-3.434293	0.664917	0.427716	-0.077746	-0.089285	0.008505
3	A	40.593380	31.169141	-8.965327	14.987658	3.871837	-1.076296	-3.135744
	B	-94.004768	-3.121490	0.591184	0.385063	-0.088260	-0.069428	0.046807
4	A	69.197328	42.593936	-1.749500	11.886439	11.368436	0.063882	-2.532366
	B	-94.059967	-2.892556	0.375632	0.562656	-0.101311	-0.137867	0.057048
5	A	38.217420	9.342198	-23.561476	-6.354079	2.093715	-0.924929	-3.669451
	B	-95.326431	-3.078919	0.274080	0.573371	-0.073677	-0.134438	0.035026
6	A	41.235765	-1.623740	-14.807304	-7.953802	2.252581	-2.483271	-4.392893
	B	-96.396988	-2.849301	0.056531	0.640160	-0.052478	-0.152187	0.031440
7	A	87.500475	37.728958	9.439757	7.979024	15.802338	8.546020	-0.302598
	B	-97.009178	-2.584683	-0.065701	0.599266	0.015779	-0.162497	0.011062
8	A	65.609077	-1.996639	1.773407	-10.931905	6.776668	-0.145174	-6.063578
	B	-98.598061	-2.506879	-0.316292	0.648415	0.048122	-0.109414	-0.026740
9	A	55.269795	-36.300311	-12.358212	-34.955968	-4.645876	-3.447424	-7.616772
	B	-99.947189	-2.197988	-0.502032	0.628253	0.052315	-0.084468	-0.063838
10	A	90.937912	-12.173224	-4.024450	-6.209097	7.734644	3.501867	4.845498
	B	-101.019524	-2.018662	-0.461378	0.479339	0.128089	-0.072979	-0.056954
11	A	35.970374	-22.567473	-11.859175	-41.227853	-15.081532	-6.725131	-5.264095
	B	-101.288704	-1.995949	-0.720020	0.477284	0.206285	-0.007499	-0.136320
12	A	42.915414	-41.251830	3.969660	-39.609522	-4.788992	-0.131423	-1.986872
	B	-102.872032	-1.262621	-0.730630	0.426863	0.075453	0.050646	-0.128807
13	A	110.542150	-80.984505	51.350854	-55.209170	11.069648	8.451405	-0.298404
	B	-105.041229	-0.151380	-0.894616	0.637029	-0.209423	0.171476	-0.137013
14	A	116.715003	-80.901086	48.219049	-43.262914	10.026372	7.031189	-2.220383
	B	-106.328239	0.384381	-0.645629	0.433913	-0.248794	0.177849	-0.071761
15	A	61.869747	-41.207033	24.478263	-21.159810	6.263315	2.085358	-1.241592
	B	-106.302246	0.181271	-0.359874	0.225811	-0.130275	0.097331	-0.037150
16	A	20.988123	-15.584491	11.624759	-8.564942	2.543398	0.868821	-0.707619
	B	-106.494965	0.109058	-0.137220	0.092715	-0.060814	0.038964	-0.013354
17	A	6.949901	-4.063347	4.080789	-2.072384	1.112890	0.581726	-0.044883
	B	-106.563614	0.040545	-0.035906	0.028195	-0.018890	0.010161	-0.004163

* The A coefficients should be multiplied by 1×10^{-4} .

4. Yoshino K, Freeman D E, Esmond I R and Parkinson W H 1983, High resolution absorption cross-section measurements and band oscillator strengths of the (1,0)-(12,0) Schumann-Runge absorption bands of O_2 . *Planet. Space Sci.* **31** 339.
5. Yoshino K, Freeman D E and Parkinson W H 1984, Atlas of the Schumann-Runge absorption bands of O_2 in the wavelength region 175-205 nm. *J phys. Chem. Ref. Data* **13**, 207.
6. Lewis B R, Berins L, Carver J H and Gibson S T 1986, Rotational variation of predissociation linewidth in the Schumann-Runge bands of $^{16}O_2$. *J Quant. Spectrosc. Radiat. Transfer* **36**, 187.
7. Allison A C, Dalgarno A and Pasachoff N W 1971 Absorption by vibrationally excited molecular oxygen in the Schumann-Runge continuum. *Planet. Space Sci.* **19**, 1463.
8. Nicolet M and Kennes R 1986, Aeronomic problems of the molecular oxygen photodissociation - I The O_2 Herzberg continuum. *Planet. Space Sci.* **34**, 1043.
9. Blake A J 1979, An atmospheric absorption model for the Schumann-Runge bands of oxygen. *J geophys. Res.* **84**, A7, 3272.
10. Lewis B R, Berins L, Carver J H and Gibson S T 1985, Decomposition of the photoabsorption continuum underlying the Schumann-Runge bands of $^{16}O_2$ -I Role of the $B^3\Sigma_u^-$ state : a new dissociation limit. *J Quant. Spectrosc. Radiat. Transfer* **33**, 627.
11. Murtagh D P 1988, The O_2 Schumann-Runge system : New calculations of photodissociation cross-sections. *Planet Space Sci.* **36**, 819.
12. Press W H, Flannery B P, Teukolsky S A and Vetterling W T *Numerical Recipes: The art of scientific computing*. Cambridge Univ. Press.

